Decoupling heterogeneity and correlation in positive multivariate continuous data: The Ge-Ga$_d$ framework with applications

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Abstract: Positive multivariate continuous data in biomedicine, risk management and other fields often exhibit right–skewness, over–dispersion, and strong positive correlation. Existing multivariate distributions/models (e.g., through additive structures or copulas) frequently suffer from rigid correlation constraints, restricted marginal dispersion, or exponential computational costs in cases of high dimensions. To address these challenges, we propose a novel *multivariate general mixture of Type I gamma* (Ge-Ga$_d$) distribution family and its corresponding mean regression framework defined on $\mathbb{R}_+^d \triangleq (0, \infty)^d$. By employing a shared latent variable multiplicative mixture mechanism, the Ge-Ga$_d$ framework naturally induces flexible positive correlations while allowing independent heterogeneous dispersion for each marginal component, thereby achieving an analytical decoupling of structural variability and individual heterogeneity. To overcome the complex integration in the log-likelihood function, we develop an efficient N-EM algorithm that collapses high-dimensional optimization into one-dimensional numerical integration, significantly enhancing computational efficiency and stability. Extensive Monte Carlo simulations and real–world applications on MEPS medical expenditure and NHANES clinical biochemical data demonstrate that the proposed Ge-Ga$_d$ models significantly outperform traditional independent and copula-based regression models in goodness-of-fit, joint probability inference, and out-of-sample predictive accuracy.

Keywords: MVGA distributions, mean regression model, N-EM algorithm, over–dispersion, positive correlation.

1. Motivation

In contemporary scientific research, researchers frequently encounter continuous multivariate data defined on the d-dimensional positive real space $\mathbb{R}_+^d \triangleq (0, \infty)^d$. Inherently, these data often present complex modeling challenges due to their strictly positive support, notable right–skewness, over–dispersion, and strong positive correlation. Applications of such data span a wide array of disciplines, including biomedical research, engineering reliability (Lawless 2011), environmental science, ecology, finance and economics (McDonald 1984). Specifically, they describe joint multi-event failure times, including medical complication onsets and mechanical component lifespans in survival analysis. And they are essential for capturing synergistic physical quantities like pollutant concentrations, analyzing mineral or biological community abundances, and modeling dynamic asset prices in financial measurement.

Statistical modeling for $\mathbb{R}_+^d$ data has developed beyond the standard Euclidean space methodology. The traditional method mainly depends on the probability structure of non-negative distribution family like lognormal, gamma, Dirichlet distributions with their multivariate generalization, and maps the data to $\mathbb{R}_+^d$ space through some scale transformations such as Box–Cox transformation and logarithmic ratio transformation to satisfy the requirements of the model (Feng *et al.* 2014). Some complex dependencies are often characterized by copula function (Aas *et al.* 2009) linking edge distribution, but the difficulties in parameter estimation, the subjectivity of model selection and the high computational costs (Oh & Patton 2016) often make it difficult for scholars to study.

In this paper, we mainly study the multivariate generalization of gamma distribution (Bownan & Shenton 1982), which is denoted as Gamma(α, β) with *probability density function* (pdf)

$$\text{Gamma}(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} \mathrm{e}^{-\beta y}, \quad y > 0,\ \alpha > 0,\ \beta > 0 \tag{1.1}$$

In existing literature, many authors have conducted a wealth of research and thinking on how to extend the gamma distribution to multivariate situation (Kotz *et al.* 2000). Cheriyan (1941) and Ramabhadran (1951) extended the gamma distribution to the case of d-dimension by *stochastic representation* (SR) $X_j = Y_0 + Y_j$ with $\{Y_j\}_{j=0}^{d} \overset{\text{ind}}{\sim} \text{Gamma}(\alpha_j, \beta)$, these SRs consider Y_0 as a shared *random variable* (r.v.), which allows positive correlation between components, and ensures that each component is gamma distribution, and the model has a relatively simple mathematical form. But at the same time, such a structure produces an identical dependency $\text{Cov}(X_j, X_{j'}) = \alpha_0/\beta$ and the same dispersion $1/\beta$ for each dimension, which makes the fitting of actual data inflexible. Krishnamoorthy & Parthasarathy (1951) defined multivariate gamma random vector $\boldsymbol{X}$ as the main diagonal element of a central Wishart distribution; that is, if $\mathbf{W}$ is a Wishart matrix, let $X_i = W_{ii}$, which makes it a natural by-product of Wishart distribution. However, the form of such construction produces extremely complex joint density, which poses serious challenges to statistical inference like parameter estimation. In the early 1990s, Mathai and Moschopoulos (1991) worked to find a general framework that could overcome the strict limitations of traditional multivariate gamma (specific correlation structure, complex density). They want to model a broader dependency pattern in reality. They defined the weighted sum vector of independent gamma variables, that is, the random vector $\boldsymbol{X} = \mathbf{A}\boldsymbol{Y}$ is a linear combination with $\{Y_j\}_{j=1}^{m} \overset{\text{ind}}{\sim} \text{Gamma}(\alpha_j, \beta_j)$. Although this makes the whole distribution have flexible correlation structure and diverse marginal distribution shapes, due to the lack of closed form joint density and complex expected variance representation, MLE inference and moment estimation for this distribution become very difficult and the solution is not unique. Semenikhine *et al.* (2018) adopted the multiplication mechanism

$$\boldsymbol{X} = (X_1, \dots, X_d)^\top \quad \text{with} \quad X_j = \frac{E_j}{\sigma_j \Lambda}, \quad j = 1, \dots, d,$$

where $\{\sigma_j\}_{j=1}^{d} > 0$, $\{E_j\}_{j=1}^{d} \overset{\text{iid}}{\sim} \text{Exponential}(1)$ and Λ follows the shifted inverse Beta distribution, whose pdf is

$$f_\Lambda(\lambda) = \frac{\lambda^{-1}(\lambda - 1)^{-\gamma}}{\Gamma(1-\gamma)\Gamma(\gamma)}, \quad \lambda > 1,\ \gamma \in (0, 1).$$

This multiplies the specific risk factor and the systemic background risk factor to construct a multivariate gamma distribution. Based on this representation, the marginal distribution $X_j \sim \text{Gamma}(\gamma, \sigma_j)$ with expectation $E(X_j) = \gamma/\sigma_j$, variance $\text{Var}(X_j) = \gamma/\sigma_j^2$ and correlation $\mathrm{Corr}(X_j, X_{j'}) = (1-\gamma)/2 \in (0, 0.5)$. At the same time, they proved that this connection mechanism belongs to the copula method with Archimedean class. In addition, Gaver (1970), Royen (1995), Dussauchoy & Berland (1975) also proposed their own multivariate gamma models, but these models all have problems such as computational difficulties, highly specialized applications or few practical applications.

In order to overcome the above problems, we propose a multivariate mixture gamma distribution in this paper. By using the *mixture representation* (MR) to replace SR often used in the traditional model, a family for multivariate mixture of *Type I gamma* (Ga) distribution (see details in Appendix B.1) is constructed to fit the positive correlation data on the d-dimensional positive real space $\mathbb{R}_+^d$. Our model is not only dependent on heterogeneity and the dispersion of each dimension, but also easy to make statistical inference due to its simple structure and mathematical form. At the same time, our distribution has strong expansibility, and covariates can be introduced to construct the mean regression model.

The rest of the paper is organized as follows. In Section 2, the multivariate general mixture of Type I gamma distribution and mean regression framework are derived, the N-EM algorithm was borrowed to calculate the *maximum likelihood estimates* (MLEs) of the model parameters. Simulation studies and comparison analyses are presented in Section 3. In Section 4, real–world applications are demonstrated through an analysis of $\mathbb{R}_+^d$ data illustrating the effectiveness of the proposed methods. Section 5 provides some conclusion and discussion of the results, and technical details and derivations are included in the Appendix. Finally, some specific cases of Ge-Ga$_d$ family with their statistical inference and numerical experiment results will be supplement in the Supplementary Materials.

2. Multivariate Ge-Ga distribution

2.1 Definition and basic properties of Ge-Ga$_d(\alpha, \boldsymbol{\mu}, \boldsymbol{\lambda})$ family

Assume that $\mathrm{Ga}(\alpha, \mu)$ denotes *Type I gamma* (Ga) distribution whose pdf is defined by (B.1). Let a positive *random variable* (r.v.) τ follow a general continuous distribution with pdf $f_\tau(\cdot|\boldsymbol{\lambda})$, where $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_m)^\top$ is an unknown parameter vector. If a d-dimensional continuous random vector $\boldsymbol{X} = (X_1, \ldots, X_d)^\top$ has the following MR:

$$\{X_j|\tau\}_{j=1}^d \overset{\text{ind}}{\sim} \text{Ga}(\alpha, \mu_j\tau) \quad \text{and} \quad \tau \sim f_\tau(\tau|\boldsymbol{\lambda}), \tag{2.1}$$

then $\boldsymbol{X}$ is said to follow the *d-dimensional general mixture of Type I gamma* (Ge-Ga$_d$) distribution, denoted by $\boldsymbol{X} \sim \text{Ge-Ga}_d(\alpha, \boldsymbol{\mu}, \boldsymbol{\lambda})$ or $\text{Ge-Ga}_d(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = \{\alpha, \boldsymbol{\mu}, \boldsymbol{\lambda}\}$ is the parameter vector and $\boldsymbol{\mu} \triangleq (\mu_1, \ldots, \mu_d)^\top$. The joint pdf of $\boldsymbol{X}$ is

$$\text{Ge-Ga}_d(\boldsymbol{x}|\alpha, \boldsymbol{\mu}, \boldsymbol{\lambda}) = \left[\prod_{j=1}^d \frac{\alpha^\alpha x_j^{\alpha-1}}{\mu_j^\alpha \Gamma(\alpha)}\right] \int_0^\infty h_*(\tau|\boldsymbol{x}, \boldsymbol{\theta}) \,\mathrm{d}\tau, \tag{2.2}$$

where

$$h_*(\tau|\boldsymbol{x}, \boldsymbol{\theta}) = \frac{f_\tau(\tau|\boldsymbol{\lambda})}{\tau^{d\alpha}} \exp\left[-\sum_{j=1}^d \frac{\alpha x_j}{\mu_j} \cdot \frac{1}{\tau}\right],$$

and the conditional density of $\tau|\boldsymbol{X}$ is

$$f_{\tau|\boldsymbol{X}}(\tau|\boldsymbol{x}) \propto \frac{f_\tau(\tau|\boldsymbol{\lambda})}{\tau^{d\alpha}} \exp\left[-\sum_{j=1}^d \frac{\alpha x_j}{\mu_j} \cdot \frac{1}{\tau}\right],$$

which can be employed to calculate the conditional expectations of $\tau|(\boldsymbol{X}, \boldsymbol{\theta})$ in the *expectation–maximization* (EM) and *data augmentation* (DA) algorithms. From the MR (2.1), for $j, j' = 1, \ldots, d$ and $j' \neq j$, we easily obtain

$$E(X_j) = \mu_j E(\tau), \quad \text{Var}(X_j) = \mu_j^2 \left\{\left(1 + \frac{1}{\alpha}\right) \text{Var}(\tau) + \frac{[E(\tau)]^2}{\alpha}\right\} \quad \text{and}$$
$$\text{Cov}(X_j, X_{j'}) = \mu_j \mu_{j'} \text{Var}(\tau).$$

2.2 MLEs of parameters via N-EM algorithm

Suppose that $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{Ge-Ga}_d(\alpha, \boldsymbol{\mu}, \boldsymbol{\lambda})$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i\}_{i=1}^n$ denote the observed data, where $\boldsymbol{x}_i$ is the observation of $\boldsymbol{X}_i$. From (2.2), we obtain the log-likelihood function of $\boldsymbol{\theta}$ as

$$\begin{aligned}\ell_*(\boldsymbol{\theta}|Y_{\text{obs}}) &= n\sum_{j=1}^d [\alpha \log \alpha - \log \Gamma(\alpha) - \alpha \log \mu_j] + \sum_{i=1}^n \sum_{j=1}^d (\alpha - 1) \log x_{ij} \\ &\quad + \sum_{i=1}^n \log \int_0^\infty h_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}) \,\mathrm{d}\tau. \end{aligned} \tag{2.3}$$

We notice that there are n complex integrals in (2.3), therefore, some traditional optimization methods like *Newton–Raphson* (NR) algorithm, Fisher scoring algorithm and *minorization–maximization* (MM) algorithm are not suitable to use. To solve these integrals, we will adopt the N-EM algorithm (Tian & Liu 2022) with the following three steps:

N-step: By normalizing $h_*(\cdot|\boldsymbol{x}_i, \boldsymbol{\theta})$, we construct a *normalizing density function* (ndf)

$$g_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}) \triangleq \frac{h_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})}{\int_0^\infty h_*(s|\boldsymbol{x}_i, \boldsymbol{\theta})\,\mathrm{d}s}, \quad \tau > 0,$$

so that $g_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})$ is also a density function defined on $\mathbb{R}_+$, where $\boldsymbol{\theta}^{(t)}$ denotes the t-th approximation of the MLE $\hat{\boldsymbol{\theta}}$.

E-step: To construct a surrogate function satisfying $Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) \leqslant \ell_*(\boldsymbol{\theta}|Y_{\mathrm{obs}})$ and $Q_*(\boldsymbol{\theta}^{(t)}|\boldsymbol{\theta}^{(t)}) = \ell_*(\boldsymbol{\theta}^{(t)}|Y_{\mathrm{obs}})$, we use the integral version of Jensen's inequality to obtain

$$\begin{aligned} Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) &= c_*^{(t)} + n\sum_{j=1}^{d}[\alpha\log\alpha - \log\Gamma(\alpha) - \alpha\log\mu_j] \\ &\quad + \sum_{i=1}^{n}\sum_{j=1}^{d}\alpha\log x_{ij} - nd\bar{d}_*^{(t)}\alpha - \sum_{i=1}^{n}\sum_{j=1}^{d}\frac{b_{i*}^{(t)}x_{ij}\alpha}{\mu_j} \\ &\quad + \sum_{i=1}^{n}\int_0^\infty \log f_\tau(\tau|\boldsymbol{\lambda})\cdot g_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \end{aligned} \tag{2.4}$$

which minorizes $\ell_*(\boldsymbol{\theta}|Y_{\mathrm{obs}})$ at $\boldsymbol{\theta} = \boldsymbol{\theta}^{(t)}$, where $c_*^{(t)}$ is a constant free from $\boldsymbol{\theta}$,

$$b_{i*}^{(t)} \triangleq \int_0^\infty \frac{1}{\tau}\cdot g_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \quad i = 1, \ldots, n,$$

$$d_{i*}^{(t)} \triangleq \int_0^\infty \log\tau\cdot g_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau \quad \text{and} \quad \bar{d}_*^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^{n} d_{i*}^{(t)}.$$

M-step: Given $\boldsymbol{\theta}^{(t)}$, by maximizing $Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})$ with respect to $\boldsymbol{\theta}$, we update $\boldsymbol{\theta}^{(t)}$ by

$$\boldsymbol{\theta}^{(t+1)} = \arg\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}).$$

Given a pre-specified small positive number δ_0, the N-, E- and M-steps are alternately repeated until

$$\left|\frac{\ell_*(\boldsymbol{\theta}^{(t+1)}|Y_{\mathrm{obs}}) - \ell_*(\boldsymbol{\theta}^{(t)}|Y_{\mathrm{obs}})}{\ell_*(\boldsymbol{\theta}^{(t)}|Y_{\mathrm{obs}})}\right| \leqslant \delta_0,$$

then we obtain the MLE $\hat{\boldsymbol{\theta}}$ of parameter vector $\boldsymbol{\theta}$. Based on (2.4), we take the first derivative of $Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})$ and solve the following system of equations:

$$\begin{aligned}
\frac{\partial Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})}{\partial \alpha} &= nd\left[C_*^{(t)} + \log\alpha - \psi(\alpha)\right] = 0, \quad \psi(\alpha) \triangleq \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}, \\
\frac{\partial Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})}{\partial \mu_j} &= -\frac{n\alpha}{\mu_j} + \frac{\alpha}{\mu_j^2}\sum_{i=1}^n b_{i*}^{(t)} x_{ij} = 0, \quad j = 1, \ldots, d, \\
\frac{\partial Q_*(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})}{\partial \lambda_k} &= \sum_{i=1}^n \int_0^\infty \frac{\partial \log f_\tau(\tau|\boldsymbol{\lambda})}{\partial \lambda_k} \cdot g_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau = 0, \quad k = 1, \ldots, m,
\end{aligned}$$

where the constant

$$C_*^{(t)} \triangleq 1 - \bar{d}_*^{(t)} - \frac{1}{d}\sum_{j=1}^d \log\mu_j + \frac{1}{nd}\sum_{i=1}^n\sum_{j=1}^d \left(\log x_{ij} - b_{i*}^{(t)} x_{ij}\mu_j^{-1}\right).$$

We obtain the $(t+1)$-th approximation of $\hat{\boldsymbol{\theta}}$ as

$$\begin{cases}
\alpha^{(t+1)} = \mathrm{sol}\left\{C_*^{(t)} + \log\alpha - \psi(\alpha) = 0\right\}, \\[2ex]
\mu_j^{(t+1)} = \dfrac{1}{n}\displaystyle\sum_{i=1}^n b_{i*}^{(t)} x_{ij}, \quad j = 1, \ldots, d, \\[2ex]
\lambda_k^{(t+1)} = \mathrm{sol}\left\{\displaystyle\sum_{i=1}^n \int_0^\infty \frac{\partial \log f_\tau(\tau|\boldsymbol{\lambda})}{\partial \lambda_k} \cdot g_*(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau = 0\right\}, \quad k = 1, \ldots, m,
\end{cases}$$

where the root of the equation $C + \log s - \psi(s) = 0$ with $s > 0$ for some constant C can be solved by the US algorithm (Li *et al.* 2026) as shown in Appendix A.1.

2.3 Mean regression model of the Ge-Ga$_d$ distribution

Assume that $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{ind}}{\sim} \text{Ge-Ga}_d(\alpha, \boldsymbol{\mu}_i, \boldsymbol{\lambda})$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i, \boldsymbol{z}_{(i)}\}_{i=1}^n$ denote the observed data, we consider the following mean regression model (McCullagh & Nelder 2019)

$$\log E(X_{ij}) = \log[\mu_{ij}E(\tau)] = \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}_j \quad \text{or} \quad \mu_{ij} = \frac{\exp(\boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}_j)}{E(\tau)}, \quad i = 1, \ldots, n, \ j = 1, \ldots, d,$$

where $\boldsymbol{z}_{(i)} = (z_{i1}, \ldots, z_{iq})^\top$ is a q-dimensional vector of covariates for the i-th subject and $\boldsymbol{\beta}_j = (\beta_{1j}, \ldots, \beta_{qj})^\top$ is the regression coefficients with $q \times d + 1 + m \leqslant n$. The log-likelihood function of parameter vector $\boldsymbol{\gamma} \triangleq \{\alpha, \boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_d, \boldsymbol{\lambda}\}$ is

$$\begin{aligned}\ell(\boldsymbol{\gamma}|Y_{\text{obs}}) &= nd[\alpha \log \alpha - \log \Gamma(\alpha)] + \sum_{i=1}^{n}\sum_{j=1}^{d}[(\alpha-1)\log x_{ij} - \alpha \log \mu_{ij}] \\ &\quad + \sum_{i=1}^{n} \log \int_0^\infty h(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}\tau,\end{aligned}$$

where $h(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) = f_\tau(\tau|\boldsymbol{\lambda}) \exp[-(\sum_{j=1}^{d} \alpha x_{ij}/\mu_{ij})/\tau]/\tau^{d\alpha}$. Similar to §2.2, we adopt the N-EM algorithm again to calculate the MLEs of $\boldsymbol{\gamma}$. The N-step of the N-EM algorithm is to compute the ndf

$$g(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) \triangleq \frac{h(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})}{\int_0^\infty h(s|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}s}, \quad \tau > 0.$$

In the E-step, we construct the surrogate function as

$$\begin{aligned}Q(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)}) &= c^{(t)} + nd[\alpha \log \alpha - \log \Gamma(\alpha)] + \sum_{i=1}^{n}\sum_{j=1}^{d}(\alpha \log x_{ij} - \alpha \log \mu_{ij}) \\ &\quad - nd\bar{d}^{(t)}\alpha - \sum_{i=1}^{n}\sum_{j=1}^{d}\frac{b_i^{(t)} x_{ij}\alpha}{\mu_{ij}} + \sum_{i=1}^{n}\int_0^\infty \log f_\tau(\tau|\boldsymbol{\lambda}) \cdot g(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau,\end{aligned}$$

where $c^{(t)}$ is a constant free from $\boldsymbol{\gamma}$,

$$b_i^{(t)} \triangleq \int_0^\infty \frac{1}{\tau} \cdot g(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, \quad i = 1, \ldots, n,$$

$$d_i^{(t)} \triangleq \int_0^\infty \log \tau \cdot g(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau \quad \text{and} \quad \bar{d}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^{n} d_i^{(t)}.$$

Finally, in the M-step, we take the first derivative of $Q(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)})$ and solve the following system of equations:

$$\begin{aligned}\frac{\partial Q(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)})}{\partial \alpha} &= n\left[C^{(t)} + \log \alpha - \psi(\alpha)\right] = 0, \\ \frac{\partial Q(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)})}{\partial \boldsymbol{\beta}_j} &= \alpha \sum_{i=1}^{n} \boldsymbol{z}_{(i)}\left(b_i^{(t)} x_{ij}\mu_{ij}^{-1} - 1\right) = \mathbf{0}_q, \quad j = 1, \ldots, d, \\ \frac{\partial Q(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)})}{\partial \lambda_k} &= \frac{\partial E(\tau)}{\partial \lambda_k} \cdot \left[\frac{nd\alpha}{E(\tau)} - \sum_{i=1}^{n}\sum_{j=1}^{d}\frac{b_i^{(t)} x_{ij}\alpha}{\mu_{ij}}\right] \\ &\quad + \sum_{i=1}^{n}\int_0^\infty \frac{\partial \log f_\tau(\tau|\boldsymbol{\lambda})}{\partial \lambda_k} \cdot g(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau = 0, \quad k = 1, \ldots, m,\end{aligned}$$

where the constant

$$C^{(t)} \triangleq 1 - \bar{d}^{(t)} + \frac{1}{nd} \sum_{i=1}^{n} \sum_{j=1}^{d} \left(\log x_{ij} - \log \mu_{ij} - b_i^{(t)} x_{ij} \mu_{ij}^{-1} \right).$$

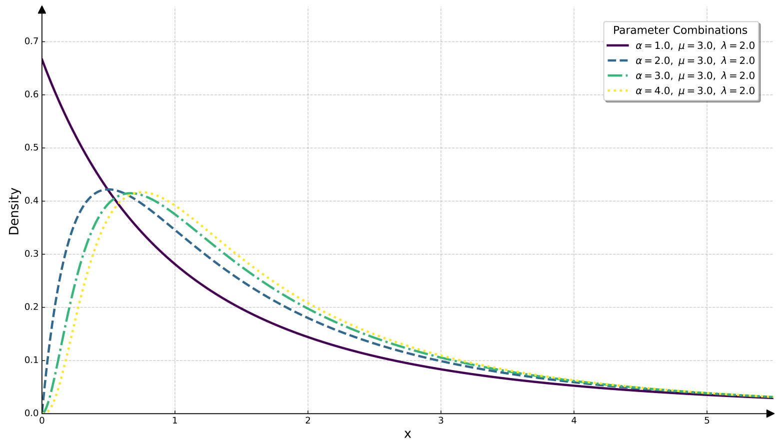
We obtain the $(t+1)$-th approximation of $\hat{\boldsymbol{\gamma}}$ as

$$\begin{cases} \alpha^{(t+1)} = \text{sol}\left\{ C^{(t)} + \log \alpha - \psi(\alpha) = 0 \right\}, \\ \boldsymbol{\beta}_j^{(t+1)} = \left(\mathbf{Z}^\top \mathbf{W}^{(t)} \mathbf{Z} \right)^{-1} \mathbf{Z}^\top \mathbf{W}^{(t)} \boldsymbol{r}_j^{(t)}, \quad j = 1, \ldots, d, \\ \lambda_k^{(t+1)} = \text{sol}\left\{ \dfrac{\partial Q(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)})}{\partial \lambda_k} = 0 \right\}, \quad k = 1, \ldots, m, \end{cases}$$

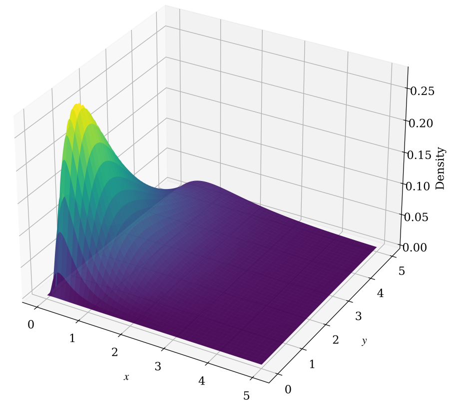
where $\mathbf{Z} = (\boldsymbol{z}_{(1)}, \ldots, \boldsymbol{z}_{(n)})^\top$ is the matrix of covariates, $\mathbf{W}^{(t)} = \text{diag}\{b_1^{(t)}, \ldots, b_n^{(t)}\}$ is the weighted matrix, $\boldsymbol{r}_j^{(t)} = (r_{1j}^{(t)}, \ldots, r_{nj}^{(t)})^\top$ with $r_{ij}^{(t)} = \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}_j^{(t)} + (x_{ij}/\mu_{ij}^{(t)} - 1/b_i^{(t)})$ is the response vector for $i = 1, \ldots, n$.

2.4 Three specific cases of the Ge-Ga$_d$ distribution family

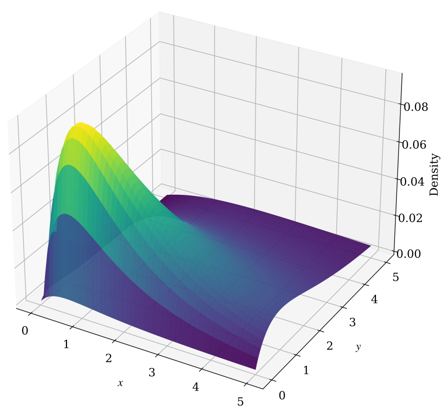
In the previous subsections, we assumed that mixture r.v. τ follows an arbitrary distribution defined on $\mathbb{R}_+$. In this subsection, we will provide three special cases of Ge-Ga$_d$ family by fixing the distribution of τ as inverse gamma, log-symmetric *generalized inverse Gaussian* (GIGau) and lognormal distributions, respectively. We define the so-called IGa-Ga$_d$, LSGIGau-Ga$_d$ and LN-Ga$_d$ distributions and study their basic properties, statistical inference together with the corresponding mean regression models (see Supplementary Materials for details). This not only expands our alternative distributions for fitting multivariate positive correlation right–skewness continuous data, but also extends the existing inverse beta distribution to multivariate formula (i.e. IGa-Ga$_d$ distribution) by our Ge-Ga$_d$ family framework, and gives a unified statistical inference method. Under different parameter combinations, the density function images of three new distributions proposed by us are shown in Figure 1.



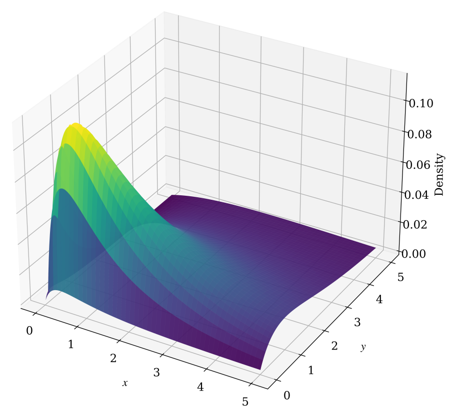
(a) IGa-Ga$_1$ (inverted Beta) distribution;



(b) IGa-Ga$_2$ distribution;



(c) LSGIGau-Ga$_2$ distribution.



(d) LN-Ga$_2$ distribution.

Figure 1: The density functions of three specific cases for Ge-Ga$_d$ distribution under various parameter combinations.

3. Simulation Studies

3.1 Accuracy of MLEs and reliability of asymptotic normal CIs

In §2.4, we propose three specific members of the Ge-Ga$_d$ distribution family and compute the MLEs via the N-EM algorithm. In addition, *confidence intervals* (CIs) of parameters can be obtained by the sandwich method (Lehmann & Casella 1998) or the bootstrap method (Diciccio & Romano 1988, Efron 1992). To assess the performance of our algorithms and methods, we conduct some numerical experiments by employing the *mean square error*

(MSE) as the criterion to evaluate the accuracy of the MLE for each parameter and apply the *coverage rate* (CR) to confirm the reliability of CIs.

In our simulations, we consider different sample sizes $n = 300(300)900$ and $K = 10{,}000$ sampling replications. We conduct some numerical experiments with following three cases:

Case (A_1): Independently generating $\{\boldsymbol{X}_i^{(k)}\}_{i=1}^n \overset{\text{ind}}{\sim} \text{IGa-Ga}_2(\alpha, \boldsymbol{\mu}_i, \lambda)$ with the true values being set as $\alpha = 2$, $\boldsymbol{\beta}_1 = (1, 2, -1)^\top$, $\boldsymbol{\beta}_2 = (0, 3, -2)^\top$ and $\text{Var}(\tau) = 0.5$;

Case (A_2): Independently generating $\{\boldsymbol{X}_i^{(k)}\}_{i=1}^n \overset{\text{ind}}{\sim} \text{LSGIGau-Ga}_2(\alpha, \boldsymbol{\mu}_i, \lambda)$ with true values being set as $\alpha = 0.5$, $\boldsymbol{\beta}_1 = (1, 0, -0.5)^\top$, $\boldsymbol{\beta}_2 = (0, 2.5, -2)^\top$ and $\text{Var}(\tau) = 1$;

Case (A_3): Independently generating $\{\boldsymbol{X}_i^{(k)}\}_{i=1}^n \overset{\text{ind}}{\sim} \text{LN-Ga}_2(\alpha, \boldsymbol{\mu}_i, \lambda)$ with the true values being set as $\alpha = 3$, $\boldsymbol{\beta}_1 = (1, 3, -2)^\top$, $\boldsymbol{\beta}_2 = (-1, 4, 0)^\top$ and $\text{Var}(\tau) = \mathrm{e} - 1$.

Given a combination of $\{n, k, \alpha, \boldsymbol{\mu}, \text{Var}(\tau)\}$ for Case (A_1), let $Y_{\text{obs}}^{(k)} = \{\boldsymbol{x}_i^{(k)}, \boldsymbol{z}_{(i)}^{(k)}\}_{i=1}^n$ denote the k-th observations with $k = 1, \ldots, K$, where $\boldsymbol{x}_i^{(k)}$ is a realization of $\boldsymbol{X}_i^{(k)}$ and the covariate $\boldsymbol{z}_{(i)}^{(k)} = (1, z_{1i}^{(k)}, z_{2i}^{(k)})^\top$ with

$$z_{1i}^{(k)} \overset{\text{iid}}{\sim} N(0,1) \quad \text{and} \quad z_{12}^{(k)} \overset{\text{iid}}{\sim} \text{Poisson}(2), \quad \text{for } i = 1, \ldots, n.$$

We can compute the MLEs of parameters via N-EM algorithm, denoted by $\{\hat{\alpha}^{(k)}, \hat{\boldsymbol{\mu}}^{(k)}, \hat{\lambda}^{(k)}\}$. Next, we calculate the *average MLE* (A-MLE) with the corresponding MSE for each parameter based on the K repetitions. The interval estimation of parameters will be obtained by sandwich method, the CR of CIs will be presented in Table 1 together with A-MLE and MSE. For Case (A_2) and (A_3), similar results are reported in Supplementary Materials and Table 2.

We found that with the increase of sample size n, the MLEs of parameters becomes more and more accurate and stable. Specifically, A-MLE is getting closer to the set true value, and MSE is getting smaller. For interval estimation, the CR is stable at about 95% with the sample size increasing, so a reliable interval estimation is obtained.

Table 1. Comparisons of A-MLE, MSE and CR for IGa-Ga_2 distribution under various sample sizes

Sample size	Parameter	A-MLE	MSE	CR
300	α	2.0256	0.0213	0.9448
	β_{11}	0.9953	0.0092	0.9480
	β_{12}	1.9999	0.0028	0.9487
	β_{13}	-0.9992	0.0014	0.9446
	β_{21}	-0.0010	0.0095	0.9439
	β_{22}	3.0000	0.0029	0.9416
	β_{23}	-2.0008	0.0015	0.9423
	$\mathrm{Var}(\tau)$	0.5104	0.0280	0.9150
600	α	2.0108	0.0102	0.9481
	β_{11}	0.9978	0.0047	0.9488
	β_{12}	2.0002	0.0014	0.9495
	β_{13}	-0.9999	0.0007	0.9464
	β_{21}	-0.0031	0.0047	0.9401
	β_{22}	2.9997	0.0014	0.9474
	β_{23}	-1.9995	0.0007	0.9404
	$\mathrm{Var}(\tau)$	0.5034	0.0120	0.9320
900	α	2.0085	0.0067	0.9468
	β_{11}	0.9987	0.0031	0.9494
	β_{12}	1.9997	0.0009	0.9490
	β_{13}	-0.9999	0.0005	0.9462
	β_{21}	-0.0010	0.0031	0.9306
	β_{22}	3.0001	0.0010	0.9455
	β_{23}	-2.0002	0.0005	0.9382
	$\mathrm{Var}(\tau)$	0.5039	0.0081	0.9427

The convergence accuracy of MLE is 10^{-8}.

Table 2. Comparisons of A-MLE and MSE for LN-Ga_2 distribution under various sample sizes

Sample size	Parameter	A-MLE	MSE
300	α	3.0433	0.0571
	$\boldsymbol{\beta}_1$	$(0.9976, 3.0002, -2.0010)^\top$	$(0.0107, 0.0034, 0.0017)^\top$
	$\boldsymbol{\beta}_2$	$(-1.0032, 3.9996, -0.0001)^\top$	$(0.0106, 0.0035, 0.0017)^\top$
	$\text{Var}(\tau)$	1.7036	0.0574
600	α	3.0211	0.0273
	$\boldsymbol{\beta}_1$	$(0.9985, 3.0001, -1.9999)^\top$	$(0.0045, 0.0014, 0.0007)^\top$
	$\boldsymbol{\beta}_2$	$(-1.0020, 4.0003, -0.0005)^\top$	$(0.0044, 0.0014, 0.0007)^\top$
	$\text{Var}(\tau)$	1.7135	0.0264
900	α	3.0141	0.0174
	$\boldsymbol{\beta}_1$	$(0.9997, 3.0003, -2.0001)^\top$	$(0.0026, 0.0008, 0.0004)^\top$
	$\boldsymbol{\beta}_2$	$(-1.0003, 4.0001, 0.0000)^\top$	$(0.0026, 0.0009, 0.0004)^\top$
	$\text{Var}(\tau)$	1.7162	0.0170

The convergence accuracy of MLE is 10^{-8}.

3.2 Model comparisons

To assess the performance of the goodness-of-fit tests among the three specific distributions (i.e., IGa-Ga_d, LSGIGau-Ga_d and LN-Ga_d) in the Ge-Ga_d distribution family and other two new distributions (denoted by IGau-Ga_d and RIGau-Ga_d, where IGau and RIGau denote *inverse Gaussian* and *reciprocal inverse Gaussian* distributions, respectively) in the Ge-Ga_d distribution family, we adopt two evaluation criteria: The *Akaike information criterion* (AIC, Akaike 1974), and the *Bayesian information criterion* (BIC, Schwarz 1978).

In our numerical studies, the sample size, the number of sampling replications are respectively set to be $n = 500$ and $K = 1{,}000$. We set the parameters $\alpha = 2$, $\boldsymbol{\mu} = (2, 1.5, 0.8)^\top$ and λ in each of the five specific distributions (denoted by λ_{IGa}, λ_{LSGIGau}, λ_{LN}, λ_{IGau}, λ_{RIGau}, respectively) can be determined by assuming that $\text{Var}(\tau) = 0.5$.

For a given $\{n, k, \alpha, \mu, \text{Var}(\tau)\}$ with $k = 1, \ldots, K$, firstly, we independently generate the samples $\{\boldsymbol{X}_i^{(k)} = \boldsymbol{x}_i^{(k)}\}_{i=1}^n$ from the mixed distribution:

$$p_1 \cdot \text{IGa-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{IGa}}) + p_2 \cdot \text{LSGIGau-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{LSGIGau}}) + p_3 \cdot \text{LN-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{LN}})$$

$$+\, p_4 \cdot \text{IGau-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{IGau}}) + p_5 \cdot \text{RIGau-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{RIGau}}).$$

where $p_s \geqslant 0$ for $s = 1, \ldots, 5$ with $\sum_{s=1}^{5} p_s = 1$. We consider the following situations for selecting $\boldsymbol{p} = (p_1, \ldots, p_5)^\top$:

Case (B$_1$): Single–component distribution: $p_s = 1$ for $s = 1, \ldots, 5$ and $p_{s'} = 0$ with $s' \neq s$, indicating that the sample is generated from each of the five specific distributions;

Case (B$_2$): Dominant two–component mixture distribution: One $p_s = 0.8$, the second $p_{s'} = 0.2$ for $s \neq s'$, and all others are zeros;

Case (B$_3$): Uniform five–component mixed distribution: $\boldsymbol{p} = 1/5 \times \mathbf{1}_5$;

Case (B$_4$): Dominant one–component mixed distribution: One specific case of Ge-Ga$_3$ distribution has a larger weight p_s and the other two have smaller weights $p_{s'}$.

Based on the generated samples $\{\boldsymbol{x}_i^{(k)}\}_{i=1}^{n}$, we can compute the MLEs $\{\hat{\alpha}^{(k)}, \hat{\boldsymbol{\mu}}^{(k)}, \hat{\lambda}^{(k)}\}$ for the five specific distributions in the Ge-Ga$_3$ distribution family via N-EM algorithm, calculate five AIC values $\{\mathsf{AIC}_1^{(k)}, \ldots, \mathsf{AIC}_5^{(k)}\}$. Then we sort AICs to obtain the corresponding five ranks $\{\mathsf{Rank}_1^{(k)}, \ldots, \mathsf{Rank}_5^{(k)}\}$, where $\mathsf{Rank}_s^{(k)}$ represents the rank of $\mathsf{AIC}_s^{(k)}$ in $\{\mathsf{AIC}_1^{(k)}, \ldots, \mathsf{AIC}_5^{(k)}\}$. Finally, the average AICs (denoted by AIC) and the average ranks (denoted by Rank) are calculated by

$$\mathsf{AIC}_s = \frac{1}{K}\sum_{k=1}^{K} \mathsf{AIC}_s^{(k)} \quad \text{and} \quad \mathsf{Rank}_s = \frac{1}{K}\sum_{k=1}^{K} \mathsf{Rank}_s^{(k)}, \quad s = 1, \ldots, 5.$$

Tables 3 presents the AICs and Ranks of five fitted distributions for Case (B$_1$) to Case (B$_3$), and the similar conclusion of Case (B$_4$) will be summarized in Supplementary Materials. From the results in Tables 3, we found that the generated Ge-Ga$_3$ distribution provides the best fitting effect, which outstanding performance is significantly better than other distributions. In general, the LSGIGau-Ga$_3$, LN-Ga$_3$ and IGau-Ga$_3$ distributions have good fitting effect in various mixed cases, means these distributions are robust to fit the complex data.

Table 3. Model comparisons based on $K = 1{,}000$ replications for Cases (B_1)—(B_3)

Mixed weight $\boldsymbol{p}$	Criterion	Fitted distribution				
		IGa-Ga$_3$	LSGIGau-Ga$_3$	LN-Ga$_3$	IGau-Ga$_3$	RIGau-Ga$_3$
$(1,0,0,0,0)^\top$	AIC	**3541.2**	3548.4	**3546.7**	3546.9	3550.0
	Rank	**4.49**	2.26	3.46	**3.62**	1.17
$(0,1,0,0,0)^\top$	AIC	3523.4	**3512.9**	3513.4	**3513.2**	3513.2
	Rank	1.15	**3.85**	3.15	**3.46**	3.39
$(0,0,1,0,0)^\top$	AIC	3259.3	**3258.0**	3258.0	**3258.0**	3258.1
	Rank	2.16	**3.28**	3.14	**3.26**	3.17
$(0,0,0,1,0)^\top$	AIC	3520.9	**3513.6**	3514.0	**3513.4**	3514.3
	Rank	1.45	**3.71**	3.10	**3.89**	2.85
$(0,0,0,0,1)^\top$	AIC	3526.3	**3511.1**	3511.8	3512.2	**3510.7**
	Rank	1.03	**3.94**	3.10	2.86	**4.07**
$\frac{1}{10}\times(9,1,0,0,0)^\top$	AIC	**3546.8**	3552.1	**3550.5**	3550.7	3553.5
	Rank	**4.14**	2.39	**3.54**	3.66	1.27
$\frac{1}{10}\times(0,9,1,0,0)^\top$	AIC	3509.6	**3500.5**	3500.7	**3500.7**	3500.8
	Rank	1.24	**3.71**	3.34	**3.40**	3.31
$\frac{1}{10}\times(0,0,9,1,0)^\top$	AIC	3294.2	3293.2	**3293.0**	**3293.2**	3293.3
	Rank	2.40	3.05	**3.58**	**3.18**	2.77
$\frac{1}{10}\times(0,0,0,9,1)^\top$	AIC	3528.2	**3520.0**	3520.5	**3520.0**	3520.7
	Rank	1.36	**3.78**	3.08	**3.78**	3.00
$\frac{1}{10}\times(1,0,0,0,9)^\top$	AIC	3529.3	**3516.2**	3516.5	3517.0	**3516.0**
	Rank	1.07	**3.81**	3.31	2.95	**3.86**
$\frac{1}{5}\times(1,1,1,1,1)^\top$	AIC	3486.0	3481.3	**3480.7**	**3481.1**	3481.8
	Rank	1.76	3.25	**3.85**	**3.37**	2.78

The convergence accuracy of MLE is 10^{-8}.

Color Red has the best fitting performance, **bolding** is the second best.

4. The Real Data Analysis

In this section, we apply the proposed distributions, mean regression models and algorithms to analyze two real data sets and compare them with the existing models to illustrate that the Ge-Ga$_d$ distributions have a more flexible and robust fitting performance.

4.1 Medical expenditure cost

The real data set considered in this study is a personal medical cost data set, which is publicly available from MEPS (https://meps.ahrq.gov/mepsweb/). Such data often show the characteristics of positive support, right–skewness and positive correlation (Mihaylova *et al.* 2011), which is suitable by using Ge-Ga$_d$ distributions to fit.

The data set used in this article is the 2023 Full Year Consolidated Data File (HC-251), which provides information collected from a nationally representative sample of the civilian non-institutionalized population of the United States for calendar year 2023. The MEPS is a follow–back survey that collected data from a sample of medical providers and pharmacies that were used by sample persons in 2023. This PUF contains the utilization and expenditure variables for several categories of health care services. It includes 11 expenditure variables (an aggregate total payments variable and 10 main component source of payment category variables with unit k dollar), usually represents the annual medical use and expenditure. After data preprocessing, we used three medical expenditure sources of 2847 samples:

SLF: Out of pocket by patients' family;

MCR: Medicare, a federal medical insurance scheme in the United States;

PRV: Private Insurance.

Sample mean vector of $(\mathsf{SLF}, \mathsf{MCR}, \mathsf{PRV})^\top$ is given by $(2.2938, 10.6730, 5.9415)^\top$ and correlation matrix is computed as

$$\mathrm{Corr}(\mathsf{SLF}, \mathsf{MCR}, \mathsf{PRV}) = \begin{pmatrix} 1.00 & 0.23 & 0.10 \\ 0.23 & 1.00 & 0.05 \\ 0.10 & 0.05 & 1.00 \end{pmatrix},$$

which shows positive correlations. We use five distributions in the Ge-Ga$_d$ distribution family compared with multivariate lognormal and multivariate gamma distributions (proposed by Cheriyan & Ramabhadran, denoted by $\mathrm{Gamma}_d^{(\mathrm{CR})}$, which can be estimated by EM algorithm, see more detailed in Appendix D) to fit the variables $(\mathsf{SLF}, \mathsf{MCR}, \mathsf{PRV})^\top$, so as to describe the distribution characteristics and morphology of individual medical expenditure. We adopt the log-likelihood value, AIC and BIC as evaluation criteria to evaluate the performance of fitting.

Table 4. Fitting performance for three individual medical expenditure (with sample mean $(2.2938, 10.6730, 5.9415)^{\top}$) of seven distributions, respectively

Distribution	Mean	Log-likelihood	AIC	BIC
IGa-Ga$_3$	$(3.5824, 14.6250, 5.7164)^{\top}$	-19433.69	38877.38	38907.15
LSGIGau-Ga$_3$	$(\mathbf{2.7680}, \mathbf{11.1203}, \mathbf{4.5438})^{\top}$	-19425.50	38861.00	38890.77
LN-Ga$_3$	$(2.7895, 11.2567, 4.5325)^{\top}$	$\mathbf{-19417.38}$	**38844.75**	**38874.52**
IGau-Ga$_3$	$(2.8388, 11.4281, 4.5943)^{\top}$	$\mathbf{-19417.41}$	**38844.82**	**38874.59**
RIGau-Ga$_3$	$(\mathbf{2.6839}, \mathbf{10.8067}, \mathbf{4.4926})^{\top}$	-19442.19	38894.37	38924.14
Lognormal$_3$	$(1.8888, 6.6033, 2.7696)^{\top}$	-19474.12	38966.25	39019.83
Gamma$_3^{(\mathrm{CR})}$	$(5.7078, 8.4420, 6.0422)^{\top}$	-20892.18	41794.36	41824.13

The convergence accuracy of MLE is 10^{-8}.

The comparison of the fitting effects of the seven distributions is shown in Table 4, from which we can see that the Ge-Ga$_3$ distribution family generally has better fitting effects. RIGau-Ga$_3$ and LSGIGau-Ga$_3$ distributions have better recovery performance on the mean value of data. LN-Ga$_3$ and IGau-Ga$_3$ distributions have the largest log-likelihood value, the smallest AIC and BIC, which means that the fitting effect of these two distributions is the best. The MLEs and CIs of parameters for LSGIGau-Ga$_3$, LN-Ga$_3$, IGau-Ga$_3$ and RIGau-Ga$_3$ distributions are presented in Table 5.

Table 5. Parameter estimations of LSGIGau-Ga$_3$, LN-Ga$_3$, IGau-Ga$_3$ and RIGau-Ga$_3$ distributions for individual medical expenditure fitting

Distribution	Parameter	MLE	Asymptotic normal		Bootstrap	
			Lower	Upper	Lower	Upper
LSGIGau-Ga$_3$	α	0.7065	0.6847	0.7284	0.6861	0.7264
	μ_1	2.7680	2.5899	2.9461	2.5821	2.9652
	μ_2	11.1203	10.3100	11.9306	10.3715	11.8785
	μ_3	4.5438	4.1428	4.9447	4.2405	4.8686
	Var(τ)	1.2943	1.1839	1.4048	1.1774	1.3938
LN-Ga$_3$	α	0.7094	0.6874	0.7314	0.6896	0.7313
	μ_1	2.7895	2.6082	2.9707	2.6383	3.0399
	μ_2	11.2567	10.4282	12.0851	10.6552	12.2676
	μ_3	4.5325	4.1332	4.9318	4.3112	4.9432
	Var(τ)	1.8238	1.5906	2.0570	1.6809	2.1932
IGau-Ga$_3$	α	0.7047	0.6830	0.7264	0.6850	0.7255
	μ_1	2.8388	2.6470	3.0306	2.6378	3.0711
	μ_2	11.4281	10.5651	12.2911	10.6249	12.2067
	μ_3	4.5943	4.1848	5.0038	4.2557	4.9192
	Var(τ)	1.6558	1.4605	1.8511	1.4625	1.8472
RIGau-Ga$_3$	α	0.7046	0.6828	0.7265	0.6830	0.7281
	μ_1	2.6839	2.5197	2.8481	2.5073	2.8346
	μ_2	10.8067	10.0428	11.5706	10.0980	11.5050
	μ_3	4.4926	4.0985	4.8866	4.2069	4.7702
	Var(τ)	1.0390	0.9751	1.1029	0.9691	1.0996

The convergence accuracy of MLE is 10^{-8}.

4.2 National health and nutrition examination survey

In order to verify the flexibility and effectiveness of the proposed Ge-Ga$_d$ mean regression model in processing high-dimensional positive data with right–skewness in the real world, this section selects the public data set of the *national health and nutrition examination survey* (NHANES) in the United States for empirical analysis. This data set can be downloaded from the official website of NHANES (https://www.cdc.gov/nchs/nhanes/) or obtained by using the nhanesA package of R language. NHANES is an authoritative large-scale continuous epidemiological survey project and its data is widely used in statistical modeling in the field of medicine and public health.

This study selects the cross-sectional data of 2017–2018 cycle, and deeply integrates the demographic characteristics, physical examination, laboratory biochemical tests and lifestyle questionnaire of the respondents. After strict data cleaning procedures such as eliminating missing values and extreme outliers, we obtain a high-quality set with 2960 samples (divide into training set and test set according to the ratio 2000 + 960) for model fitting. In clinical blood tests, biochemical indicators often show the feature of strictly positive, over–dispersion and high positive correlation among different dimensions, which is highly consistent with the core assumptions of the Ge-Ga$_d$ family, and provides an ideal test scenario for the application of its regression framework.

In the setting of model variables, we select two core biochemical enzymes for clinical evaluation of hepatobiliary function as multivariate dependent variables, namely *alanine aminotransferase* (ALT) and *aspartate aminotransferase* (AST). From the perspective of biological mechanism, ALT and AST mainly exist in the interior of hepatocytes. When the liver is stimulated by inflammation, fat accumulation or chemical substances, the permeability of cell membrane increases, leading to a large number of escape of these two enzymes into the blood. Therefore, the activity levels of these two biochemical enzymes are jointly driven by the underlying metabolic homeostasis and liver and gallbladder health (i.e., the mixture variable τ) set in our model, showing a strong synergistic change and positive correlation in statistics:

$$\mathrm{Corr}(\mathsf{ALT}, \mathsf{AST}) = \begin{pmatrix} 1.00 & 0.77 \\ 0.77 & 1.00 \end{pmatrix}.$$

In order to further explore the external risk drivers of abnormal hepatobiliary enzymes, we construct a covariate matrix including age (Age), gender (Gender), body mass index (BMI), glycosylated hemoglobin (HbA1c), systolic blood pressure (SysBP) and average daily alcohol consumption (Alcohol). These covariates comprehensively cover basic demographic characteristics, metabolic syndrome indications and bad living habits (Clark *et al.* 2003), enabling us to accurately quantify the combined effects of various risk factors on multiple hepatobiliary functions through the Ge-Ga$_d$ mean regression models

$$\begin{aligned}
(\mathsf{ALT}, \mathsf{AST})^\top &\overset{\text{ind}}{\sim} \text{Ge-Ga}_2(\alpha, \boldsymbol{\mu}_i, \lambda) \quad \text{and} \\
\log \mu_{ij} &= \beta_{1j} + \mathsf{Age}_i \times \beta_{2j} + \mathsf{Gender}_i \times \beta_{3j} + \mathsf{BMI}_i \times \beta_{4j} + \mathsf{HbA1c}_i \times \beta_{5j} \\
&\quad + \mathsf{SysBP}_i \times \beta_{6j} + \mathsf{Alcohol}_i \times \beta_{7j} \quad i = 1, \ldots, 2000, \ j = 1, 2.
\end{aligned}$$

This section aims to accurately quantify the impact of these basic demographic characteristics, metabolic indicators and living habits on the activity of multiple biochemical enzymes. By comparing with the mean regression models of independent gamma, independent Inverse Gaussian, gamma with Gaussian Copula and inverse Gaussian with Gaussian Copula, we present the fitting performance on training set in Table 6 below.

Table 6. Fitting performance for three biological enzyme activities of eight regression models on training set, respectively

Model	Log-likelihood	AIC	BIC
IGa-Ga$_2$	**−5019.636**	**10071.27**	**10163.59**
LSGIGau-Ga$_2$	−5182.688	10397.38	10489.69
LN-Ga$_2$	**−5153.480**	**10338.96**	**10431.28**
IGau-Ga$_2$	−5170.767	10373.53	10465.85
RIGau-Ga$_2$	−5194.611	10421.22	10513.54
Independent gamma	−6604.787	13241.57	13333.89
Independent inverse Gaussian	−6097.840	12227.68	12320.00
Copula with gamma	−5455.643	10947.29	11051.14
Copula with inverse Gaussian	−5536.230	11108.46	11212.32

The convergence accuracy of MLE is 10^{-8}.

Table 7 presents the out-of-sample performance of the proposed multivariate mean regression models and four baseline models on the test set. In terms of overall goodness-of-fit, the five proposed Ge-Ga$_2$ family models demonstrate an overwhelming advantage over traditional models across log-likelihood, AIC, and BIC value. Among them, the IGa-Ga$_2$ model performs the best, achieving the highest test log-likelihood and the minimum information criteria. In contrast, the independent Gamma and independent Inverse Gaussian (IGau) models, which ignore the positive correlation between dimensions, exhibit the poorest fitting performance.

Table 7. Fitting performance for three biological enzyme activities of eight regression models on test set, respectively

Model	Log-likelihood	AIC	BIC	MAE	
				ALT	AST
IGa-Ga$_2$	**−1258.220**	**2548.441**	**2618.577**	**0.9424**	0.6667
LSGIGau-Ga$_2$	−1292.560	2617.121	2687.257	0.9500	0.6797
LN-Ga$_2$	**−1287.055**	**2606.110**	**2676.246**	**0.9473**	0.6744
IGau-Ga$_2$	−1290.258	2612.517	2682.653	0.9504	0.6805
RIGau-Ga$_2$	−1294.876	2621.753	2691.889	0.9498	0.6791
Independent Ga	−1766.945	3565.890	3636.026	0.9729	**0.6631**
Independent IGau	−1653.049	3338.098	3408.234	0.9777	**0.6641**
Copula with gamma	−1488.002	3012.003	3090.907	0.9729	**0.6631**
Copula with inverse Gaussian	−1389.523	2815.045	2893.948	0.9777	**0.6641**

Mean absolute error can be computed as $\mathsf{MAE} = \frac{1}{n'}\sum_{i=1}^{n'} |x_i - \hat{\mu}_i|$.

The convergence accuracy of MLE is 10^{-8}.

Although the introduction of the Gaussian Copula function effectively captures this correlation through an external link and substantially improves the test likelihood, it still lags behind the Ge-Ga$_2$ family. This compellingly demonstrates the unparalleled superiority of the Ge-Ga$_2$ distribution, constructed via an internal latent mixture mechanism, in characterizing the joint distribution of high-dimensional, right–skewed clinical data.

Furthermore, regarding the prediction accuracy of the conditional mean (MAE), the Copula models maintain identical MAE values to the independent GLMs, as theoretically expected. Meanwhile, the Ge-Ga$_2$ family models not only achieve lower prediction errors on the ALT indicator but also perform highly comparably to the baseline models on AST. Overall, the proposed Ge-Ga$_2$ regression framework achieves a significant leap in multivariate joint probability inference capabilities while maintaining excellent marginal point prediction accuracy.

5. Discussions

In this paper, we proposed a novel multivariate general mixture of Ga (Ge-Ga$_d$) distribution family and its corresponding mean regression framework to address the prevalent modeling challenges of right–skewness, severe over–dispersion, and high-dimensional positive synergistic coupling in $\mathbb{R}_+^d$ continuous data. Departing from traditional additive multivariate constructions and copula-based stitching methods, we innovatively introduced a multiplicative mixture mechanism driven by a shared positive latent variable τ. This theoretical architecture naturally induces flexible positive correlations among dimensions at the foundational level while allowing each marginal component to retain its heterogeneous dispersion characteristics. Consequently, it achieves an analytical decoupling of population–level structural variability and individual–level random heterogeneity. To overcome the complex high-dimensional integration bottlenecks typical of multivariate mixture models, we developed an efficient N-EM algorithm. This approach elegantly collapses the optimization over high-dimensional parameter spaces into one-dimensional numerical integration with respect to the latent variable, significantly enhancing computational efficiency and numerical stability. Extensive numerical simulations and empirical analyses based on MEPS and NHANES datasets compellingly demonstrate that the proposed Ge-Ga$_d$ regression models substantially outperform traditional independent distribution models and Copula regression models in terms of information criteria, joint probability inference, and out-of-sample predictive accuracy.

Although the Ge-Ga$_d$ framework exhibits outstanding performance in handling complex multivariate positive data, several highly promising directions warrant further investigation in future research:

First, transitioning toward a fully data-driven, non-parametric adaptive distribution paradigm. The current model construction relies on the prior parametric assumption that the mixture latent variable τ follows a specific distribution family (e.g., the generalized inverse Gaussian or log-normal distributions). When confronting extremely complex and unknown noise in the real world, such parametric assumptions inevitably harbor the risk of model misspecification. Future research could abandon specific distributional assumptions to establish a *non-parametric multivariate mixture of Ga* (NPMGA$_d$) distribution theory. By introducing Gauss–Laguerre quadrature nodes for histogram-based discretization, we assume that the mixture variable τ follows:

$$p(\tau = \tau_j) = p_j, \quad j = 1, \dots, M \quad \text{or} \quad F_\tau(\tau|\boldsymbol{p}) = \sum_{j=1}^{M} p_j \cdot I(\tau \geqslant \tau_j),$$

where $\boldsymbol{p} = (p_1, \dots, p_M)^\top$ an unknown parameter vector of weights satisfying $\boldsymbol{p} \in \mathbb{T}_M$ with

$$\mathbb{T}_M \triangleq \left\{\boldsymbol{p} = (p_1, \dots, p_M)^\top \colon\, 0 \leqslant p_j \leqslant 1,\ j = 1, \dots, M,\ \boldsymbol{p}^\top \mathbf{1}_M = 1\right\},$$

and $0 < \tau_1 < \cdots < \tau_M$ is a partition of interval $\mathbb{R}_+$. Deeply integrating Sieve regularization theory and *false discovery rate* (FDR) mechanisms to suppress high-dimensional over–fitting, and employing *minorization–maximization* (MM) algorithms, the model could adaptively approximate any arbitrarily complex latent heterogeneity structure. This would provide a highly robust and trustworthy predictive foundation with excellent *out-of-distribution* (OOD) generalization capabilities for high-risk applications.

Second, in complex scenarios of multimodal joint modeling, constructing a unified probabilistic mechanism capable of simultaneously accommodating and precisely characterizing both positive correlation and negative correlation structures remains a challenging yet significant open problem in multivariate statistical distribution theory, as the current shared latent variable mechanism predominantly induces positive dependence.

Supplementary materials

In the supplement, we provide three special cases of Ge-Ga$_d$ distribution family in detail, specifically introduce their definitions, basic properties, statistical inference and corresponding mean regression models. At the same time, we added some US algorithms we constructed and the related distributions we need to use. Some numerical experiment results will also be presented in supplement. Interested readers can see the supplement in https://github.com/YuanfanZhao/MVGA/blob/main/Thesis/Supplementary_Materials.pdf.

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A. Some technical derivations

A.1 The root of $C + \log s - \psi(s) = 0$ with $s > 0$ via a US algorithm

The aim of this appendix is to solve the root of the equation $g(s) \triangleq C + \log s - \psi(s) = 0$ with $s > 0$ by employing a US algorithm. We know the digamma function $\psi(\cdot)$ is defined as

$$\psi(s) \triangleq \frac{\Gamma'(s)}{\Gamma(s)} = -\gamma_0 + \sum_{m=0}^{\infty} \left(\frac{1}{m+1} - \frac{1}{m+s} \right),$$

where $\gamma_0 = \lim_{n\to\infty} \left(\sum_{m=1}^{n} m^{-1} - \log n \right) \approx 0.5772$ is the Euler–Mascheroni constant. Since

$$\begin{aligned} g'(s) &= \frac{1}{s} - \psi'(s) > -\psi'(s) = -\sum_{m=0}^{\infty} \frac{1}{(m+s)^2} = -\frac{1}{s^2} - \sum_{m=1}^{\infty} \frac{1}{(m+s)^2} \\ &> -\frac{1}{s^2} - \sum_{m=1}^{\infty} \frac{1}{m^2} = -\frac{1}{s^2} - \frac{\pi^2}{6} \triangleq b(s), \end{aligned}$$

the US iteration for calculating $s^{(t+1)}$ is

$$\begin{aligned} s^{(t+1)} &= \mathrm{sol}\left\{ g(s^{(t)}) + \int_{s^{(t)}}^{s} b(z)\,\mathrm{d}z = 0,\ \forall\, s, s^{(t)} > 0 \right\} \\ &= \mathrm{sol}\left\{ g(s^{(t)}) + \frac{1}{s} - \frac{\pi^2}{6}s - \frac{1}{s^{(t)}} + \frac{\pi^2}{6}s^{(t)} = 0,\ \forall\, s, s^{(t)} > 0 \right\} \\ &= \mathrm{sol}\left\{ \frac{\pi^2}{6}s^2 - q^{(t)}s - 1 = 0,\ \forall\, s, s^{(t)} > 0 \right\} \\ &= \frac{q^{(t)} + \sqrt{q^{(t)2} + 2\pi^2/3}}{\pi^2/3}, \end{aligned}$$

where

$$q^{(t)} = C + \log s^{(t)} - \psi(s^{(t)}) + \frac{\pi^2 s^{(t)}}{6} - \frac{1}{s^{(t)}}.$$

B. Type I gamma distribution and mean regression model

B.1 Type I gamma distribution

The pdf of the r.v. $Y \sim \mathrm{Gamma}(\alpha, \beta)$ is given by (1.1), where $\alpha\,(>0)$ is called the shape parameter, $\beta\,(>0)$ the rate parameter, and $\mu \triangleq \alpha/\beta\,(>0)$ the mean parameter. If we adopt parameters $\{\alpha, \mu\}$ instead of $\{\alpha, \beta\}$, then the reparametrized gamma distribution is called *Type I gamma* (Ga) distribution, denoted by $X \sim \mathrm{Ga}(\alpha, \mu)$ with pdf

$$\mathrm{Ga}(x|\alpha, \mu) = \frac{\alpha^\alpha}{\mu^\alpha \Gamma(\alpha)} x^{\alpha-1} \exp\left(-\frac{\alpha x}{\mu}\right), \quad x > 0. \tag{B.1}$$

The moment generating function and the k-th moments of X are given by

$$\begin{aligned} M_X(t) &= \left(\frac{\alpha}{\alpha - \mu t}\right)^\alpha, \quad t < \frac{\alpha}{\mu} \quad \text{and} \\ E(X^k) &= \left(\frac{\mu}{\alpha}\right)^k \prod_{j=0}^{k-1}(\alpha + j), \quad k \in \mathbb{N}_+ \triangleq \{1, 2, \ldots, \infty\}. \end{aligned}$$

Thus, we can compute the expectation, variance and *coefficient of variation* (CV) of X as

$$E(X) = \mu, \quad \mathrm{Var}(X) = \frac{\mu^2}{\alpha} \quad \text{and} \quad \mathrm{CV}(X) \triangleq \frac{\sqrt{\mathrm{Var}(X)}}{E(X)} = \sqrt{\frac{1}{\alpha}},$$

respectively. Thus, we say μ the mean parameter of Ga distribution and α the shaper (or CV) parameter. In addition, Ga distribution has the following properties:

Theorem 1 (Some properties of Ga distribution)

1° If $Y \sim \mathrm{Ga}(\alpha, \mu)$, then $Y^{-1} \sim \mathrm{IGamma}(\alpha, \alpha/\mu)$, whose pdf is defined by

$$\mathrm{IGa}(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{-\alpha-1} \mathrm{e}^{-\beta/y}, \quad y > 0;$$

2° $\mathrm{Ga}(1, \mu) = \mathrm{Exponential}(1/\mu)$;

3° $\mathrm{Ga}(\nu/2, \nu) = \chi^2(\nu)$;

4° If $Y \sim \mathrm{Ga}(\alpha, \mu)$, then $cY \sim \mathrm{Ga}(\alpha, c\mu)$ for any constant $c > 0$. $\|$

B.2 Type I gamma mean regression model and iterative reweighted least squares approach

Suppose that $\{Y_i\}_{i=1}^n \stackrel{\text{ind}}{\sim} \text{Ga}(\alpha, \mu_i)$ with common shape parameter α and each mean parameter μ_i, we consider the following Type I gamma mean regression model

$$\log \mu_i = \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta} \quad \text{or} \quad \mu_i = \exp\left(\boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}\right),$$

where $\boldsymbol{z}_{(i)}^\top = (z_{i1}, \ldots, z_{ip})^\top$ is a p-dimensional vector of covariates for the i-th subject and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^\top$ is the regression coefficients with $p+1 \leqslant n$. The log-likelihood function of $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}^\top)^\top$ is

$$\ell(\boldsymbol{\theta}|Y_{\text{obs}}) = \text{constant} + n\left[\left(\log\alpha + \frac{1}{n}\sum_{i=1}^n \log y_i\right)\alpha - \log\Gamma(\alpha)\right] - \alpha\sum_{i=1}^n\left(\log\mu_i + \frac{y_i}{\mu_i}\right),$$

where $Y_{\text{obs}} = \{y_i, \boldsymbol{z}_i\}_{i=1}^n$. For the shape parameter α, we easily obtain

$$\frac{\partial\ell(\boldsymbol{\theta}|Y_{\text{obs}})}{\partial\alpha} = n\left[1 + \frac{1}{n}\log y_i - \frac{1}{n}\sum_{i=1}^n\left(\log\mu_i + \frac{y_i}{\mu_i}\right) + \log\alpha - \psi(\alpha)\right] = 0,$$

which can be solved by US algorithm in Appendix A.1. And for the regression coefficients $\boldsymbol{\beta}$, by taking the first-order and second-order partial derivatives of the log-likelihood function, we obtain

$$\begin{aligned}
\nabla\ell(\boldsymbol{\beta}) &\triangleq \frac{\partial\ell(\boldsymbol{\theta}|Y_{\text{obs}})}{\partial\boldsymbol{\beta}} = \alpha\sum_{i=1}^n\left(\frac{y_i-\mu_i}{\mu_i}\right)\boldsymbol{z}_{(i)} = \mathbf{Z}^\top\mathbf{W}\boldsymbol{u}, \\
\mathbf{I}(\boldsymbol{\beta}|Y_{\text{obs}}) &\triangleq -\frac{\partial^2\ell(\boldsymbol{\theta}|Y_{\text{obs}})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top} = \alpha\sum_{i=1}^n\frac{y_i\boldsymbol{z}_{(i)}\boldsymbol{z}_{(i)}^\top}{\mu_i} = \mathbf{Z}^\top\mathbf{W}\mathbf{Y}\mathbf{M}^{-1}\mathbf{Z},
\end{aligned}$$

where $\boldsymbol{u} = (u_1, \ldots, u_p)^\top$ with $u_i = (y_i - \mu_i)/\mu_i$, $\mathbf{Y} = \text{diag}\{y_1, \ldots, y_n\}$, $\mathbf{M} = \text{diag}\{\mu_1, \ldots, \mu_n\}$, $\mathbf{W} = \alpha\cdot\mathbf{I}_n$ is the weight matrix and $\mathbf{Z} = (\boldsymbol{z}_{(1)}, \ldots, \boldsymbol{z}_{(n)})^\top$ is the covariate matrix. Therefore, we obtain the Fisher information matrix of $\boldsymbol{\beta}$ is $\mathbf{J}(\boldsymbol{\beta}) = E[\mathbf{I}(\boldsymbol{\beta}|Y_{\text{obs}})] = \mathbf{Z}^\top\mathbf{W}\mathbf{Z}$. Based on the Fisher scoring algorithm, we have

$$\begin{aligned}
\boldsymbol{\beta}^{(t+1)} &= \boldsymbol{\beta}^{(t)} + \mathbf{J}^{-1}\left(\boldsymbol{\beta}^{(t)}\right)\nabla\ell\left(\boldsymbol{\beta}^{(t)}\right) = \boldsymbol{\beta}^{(t)} + \left(\mathbf{Z}^\top\mathbf{W}\mathbf{Z}\right)^{-1}\mathbf{Z}^\top\mathbf{W}\boldsymbol{u}^{(t)} \\
&= \left(\mathbf{Z}^\top\mathbf{W}\mathbf{Z}\right)^{-1}\mathbf{Z}^\top\mathbf{W}\left(\mathbf{Z}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}\right) \\
&= \left(\mathbf{Z}^\top\mathbf{W}\mathbf{Z}\right)^{-1}\mathbf{Z}^\top\mathbf{W}\boldsymbol{r}^{(t)},
\end{aligned}$$

where $\boldsymbol{r} = \mathbf{Z}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}$ is the response vector and $\mathbf{W}$ is a diagonal weight matrix, which is the form of *iterative reweighted least squares* (IRLS).

C. Cheriyan & Ramabhadran's multivariate gamma distribution

Cheriyan & Ramabhadran (1951) extended the gamma distribution to d-dimension random vector $\boldsymbol{X} = (X_1, \dots, X_d)^\top$ via the following SRs:

$$X_j \overset{\mathrm{d}}{=} Y_0 + Y_j, \quad j = 1, \dots d, \tag{C.1}$$

where $\{Y_j\}_{j=0}^d \overset{\text{ind}}{\sim} \text{Gamma}(\alpha_j, \beta)$, denoted by $\boldsymbol{X} \sim \text{Gamma}_d^{(\text{CR})}(\boldsymbol{\theta})$ with parameter vector $\boldsymbol{\theta} = (\alpha_0, \alpha_1, \dots, \alpha_d, \beta)^\top$. These SRs consider Y_0 as a shared r.v. through the structure of addition, which allows positive correlation between components, and ensures that each component has a univariate gamma distribution, and the model has a relatively simple mathematical form.

Based on the above SRs, we easily obtain the marginal distribution $X_j \sim \text{Gamma}(\alpha_0 + \alpha_j, \beta)$, the expectation and variance of X_j are

$$E(X_j) = \frac{\alpha_0 + \alpha_j}{\beta}, \quad \text{and} \quad \text{Var}(X_j) = \frac{\alpha_0 + \alpha_j}{\beta^2}, \quad j = 1, \dots .d.$$

For the correlation between components with $j \neq j'$, we have

$$\text{Cov}(X_j, X_{j'}) = \text{Var}(Y_0) = \frac{\alpha_0}{\beta^2} \quad \text{and} \quad \text{Corr}(X_j, X_{j'}) = \frac{\alpha_0}{\sqrt{(\alpha_0 + \alpha_j)(\alpha_0 + \alpha_{j'})}},$$

showing positive correlations introduced through the shared r.v. Y_0 (controlled by α_0).

To calculate the MLEs of parameters in this distribution, we adopt the EM algorithm. Suppose that $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{Gamma}_d^{(\text{CR})}(\boldsymbol{\theta})$ and the observed data $Y_{\text{obs}} = \{\boldsymbol{x}_i\}_{i=1}^n$. For a fixed $i \in \{1, \dots, n\}$, corresponding to $\boldsymbol{X}_i = (X_{i1}, \dots, X_{ij}, \dots, X_{id})^\top$, based on (C.1), we introduce latent variables $\{Y_{i0}, Y_{ij}\}_{j=1}^d$ such that

$$X_{ij} = Y_{i0} + Y_{ij}, \quad j = 1, \dots d,$$

where $Y_{10}, \dots, Y_{n0} \overset{\text{iid}}{\sim} \text{Gamma}(\alpha_0, \beta)$, $\{Y_{1j}, \dots, Y_{nj}\}_{j=1}^d \overset{\text{ind}}{\sim} \text{Gamma}(\alpha_j, \beta)$ and they are independent. And we obtained the missing data $Y_{\text{mis}} = \{y_{i0}, y_{i1}, \dots, y_{id}\}_{i=1}^n$ with $y_{ij} = x_{ij} - y_{i0}$

so that $Y_{\text{mis}} = \{y_{i0}\}_{i=1}^n$, and the complete data $Y_{\text{com}} = \{Y_{\text{obs}}, Y_{\text{mis}}\} = \{x_{i1}, \ldots, x_{id}, y_{i0}\}_{i=1}^n = \{y_{i0}, y_{i1}, \ldots, y_{id}\}_{i=1}^n$. Therefore, we obtain the complete data log-likelihood function

$$\begin{aligned}\ell(\boldsymbol{\theta}|Y_{\text{com}}) &= \sum_{i=1}^n \Bigg\{\alpha_0 \log\beta - \log\Gamma(\alpha_0) + (\alpha_0 - 1)\log y_{i0} - \beta y_{i0} \\ &\quad + \sum_{j=1}^d [\alpha_j \log\beta - \log\Gamma(\alpha_j) + (\alpha_j - 1)\log(x_{ij} - y_{i0}) - \beta(x_{ij} - y_{i0})]\Bigg\}.\end{aligned}$$

For the i-th sample, based on Bayes formula, we obtain the conditional predictive distribution

$$f(y_{i0}|\boldsymbol{x}_i, \boldsymbol{\theta}) \propto y_{i0}^{\alpha_0 - 1}\left[\prod_{j=1}^d (x_{ij} - y_{i0})^{\alpha_j - 1}\right]\mathrm{e}^{(d-1)\beta y_{i0}}, \quad 0 < y_{i0} < \min\{x_{i1}, \ldots, x_{id}\}.$$

The conditional expectation can be calculated as

$$E(Y_{i0}|\boldsymbol{x}_i, \boldsymbol{\theta}) = \int_0^{m_i} s \cdot f(s|\boldsymbol{x}_i, \boldsymbol{\theta})\,\mathrm{d}s, \quad E(\log Y_{i0}|\boldsymbol{x}_i, \boldsymbol{\theta}) = \int_0^{m_i} \log s \cdot f(s|\boldsymbol{x}_i, \boldsymbol{\theta})\,\mathrm{d}s \quad \text{and}$$

$$E[\log(X_{ij} - Y_{i0})|\boldsymbol{x}_i, \boldsymbol{\theta}] = \int_0^{m_i} \log(x_{ij} - s) \cdot f(s|\boldsymbol{x}_i, \boldsymbol{\theta})\,\mathrm{d}s, \quad j = 1, \ldots, d.$$

By taking the first derivative of $\ell(\boldsymbol{\theta}|Y_{\text{com}})$ and solve the following system of equations:

$$\begin{aligned}\frac{\partial\ell(\boldsymbol{\theta}|Y_{\text{com}})}{\partial\alpha_0} &= n\log\beta - n\psi(\alpha_0) + \sum_{i=1}^n \log y_{i0} = 0, \\ \frac{\partial\ell(\boldsymbol{\theta}|Y_{\text{com}})}{\partial\alpha_j} &= n\log\beta - n\psi(\alpha_j) + \sum_{i=1}^n \log(x_{ij} - y_{i0}) = 0, \\ \frac{\partial\ell(\boldsymbol{\theta}|Y_{\text{com}})}{\partial\beta} &= \frac{n\sum_{j=0}^j \alpha_j}{\beta} - \sum_{i=1}^n\sum_{j=1}^d x_{ij} + (d-1)\sum_{i=1}^n y_{i0} = 0.\end{aligned}$$

We obtain the $(t+1)$-th approximation of $\hat{\boldsymbol{\theta}}$ as

$$\begin{cases} \alpha_0^{(t+1)} = \text{sol}\left\{\log\beta^{(t)} + \dfrac{1}{n}\displaystyle\sum_{i=1}^n \log y_{i0} - \psi(\alpha_0) = 0\right\}, \\ \alpha_j^{(t+1)} = \left\{\log\beta^{(t)} + \dfrac{1}{n}\displaystyle\sum_{i=1}^n \log(x_{ij} - y_{i0}) - \psi(\alpha_j) = 0\right\}, \quad j = 1, \ldots, d, \\ \beta^{(t+1)} = \dfrac{n\sum_{j=0}^j \alpha_j^{(t+1)}}{\sum_{i=1}^n \left[\sum_{j=1}^d x_{ij} - (d-1)y_{i0}\right]}, \end{cases}$$

where the equation $C - \psi(s) = 0$ for $s > 0$ can be solved by US algorithm and we update $\hat{\boldsymbol{\theta}}$ by replacing y_{i0}, $\log y_{i0}$, $\log(x_{ij} - y_{i0})$ with $E(Y_{i0}|\boldsymbol{x}_i, \boldsymbol{\theta})$, $E(\log Y_{i0}|\boldsymbol{x}_i, \boldsymbol{\theta})$ and $E[\log(X_{ij} - Y_{i0})]$, respectively.